

MonteCarlo: From Classic to Quantum

I. Classical MarkovChain Monte Carlo Algorithms

Yongkang Xu

Center of Correlated Matter, Zhejiang University

May 14, 2026

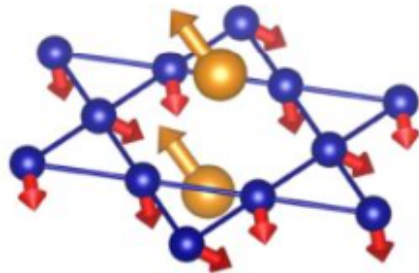
Outline

- 1 Motivation: TbCo_2 and the Kondo-Ising Model
- 2 Monte Carlo Fundamentals
- 3 Classical Algorithms: Metropolis, SW, and Wolff
- 4 Wolff Algorithm on Infinite Lattice
- 5 Loop and Worm Algorithms for Quantum Systems
- 6 Summary and Outlook

TbCo₂: A Ferrimagnetic Laves Phase Compound

- TbCo₂ crystallizes in the **cubic C15 Laves phase** structure
- Co atoms form a network of corner-sharing tetrahedra
- Tb occupies the larger interstitial sites
- Ferrimagnetic order: Tb moments ($\sim 9 \mu_B$) and Co moments ($\sim 1 \mu_B$) couple *antiparallel*

Ref: Applied Physics Letters **121**, 232404 (2022)

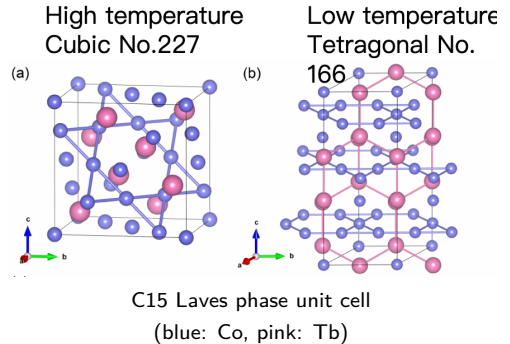


$$0 < \mu_0 H < \mu_0 H_s$$

Non-collinear ferrimagnetic structure
under intermediate field $0 < \mu_0 H < \mu_0 H_s$

Crystal Structure: Co Kagome Sublattice

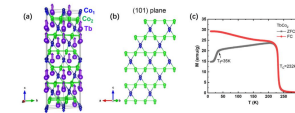
- In certain crystallographic planes, Co atoms form a **kagome lattice**
- Tb atoms sit on a separate diamond-cubic sublattice
- The interplay of two magnetic sublattices gives rise to rich physics:
 - Anomalous Hall effect
 - Topological Hall effect
 - Non-coplanar spin textures



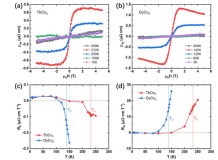
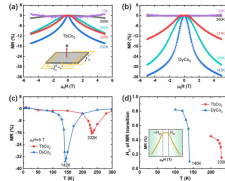
Anomalous Transport Properties of TbCo₂

- Non-linear Hall resistivity ρ_{xy} — characteristic of **Anomalous Hall Effect (AHE)**
- **Topological Hall Effect (THE)** signal observed near $T_C = 232$ K
- Sharp negative magnetoresistance peak at T_C
- These signatures point to a non-trivial spin-charge coupling mechanism

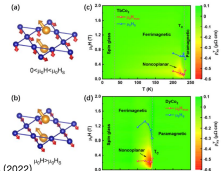
□ Polycrystalline samples results



- Plateau-like magnetoresistance



Appl. Phys. Lett. 121, 232404 (2022)

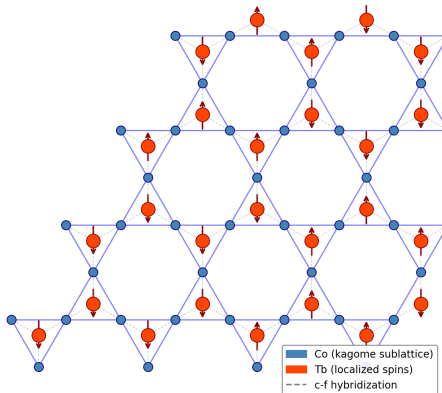


Transport measurements on polycrystalline TbCo₂
(Appl. Phys. Lett. **121**, 232404)

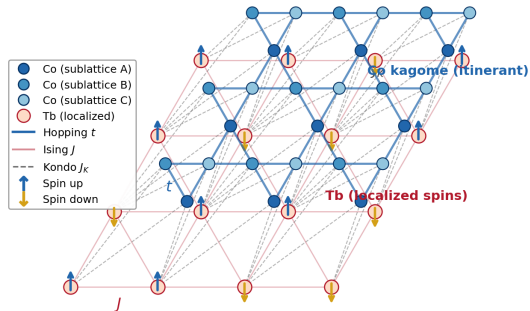
From TbCo₂ to the Kondo-Ising Model

Kondo-Ising Model on Kagome Lattice

$$H = -t \sum_{\langle ij \rangle \sigma} c_{i\sigma}^\dagger c_{j\sigma} + J_K \sum_i S_i^z S_i^z - J \sum_{\langle ij \rangle} S_i^z S_j^z$$



2D kagome lattice view



$$H = -t \sum_{\langle ij \rangle \sigma} c_{i\sigma}^\dagger c_{j\sigma} + J_K \sum_i S_i^z S_i^z - J \sum_{\langle ij \rangle} S_i^z S_j^z$$

Isometric bilayer view

Kondo-Ising Model: Hamiltonian

Kondo-Ising Hamiltonian

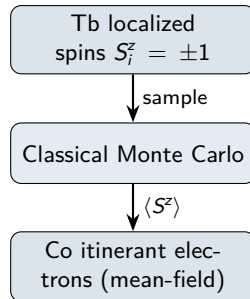
$$H = \underbrace{-t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma}}_{\text{itinerant electrons (Co)}} + \underbrace{J_K \sum_i S_i^z s_i^z}_{\text{Kondo coupling}} - \underbrace{J \sum_{\langle ij \rangle} S_i^z S_j^z}_{\text{Ising interaction (Tb)}}$$

- **Assumption:** c–f hybridization is weak $\rightarrow$ project onto Ising anisotropy limit
- The model separates naturally into two parts:
 - ① **Itinerant electron part (Co):** temperature enters only through Fermi–Dirac smearing
 - ② **Localized spin part (Tb):** treated as a classical Ising system via **Monte Carlo**
- The localized spins act as an effective **mean field** on itinerant electrons

Refs: W.-W. Yang *et al.*, PRB **100**, 045148 (2019); PRB **102**, 195141 (2020); PRB **104**, 165146 (2021)

Roadmap: Why Classical Monte Carlo?

- The Kondo-Ising model requires **thermal sampling** of the localized spin configuration S_i^z at each temperature
- This is a **classical** Monte Carlo problem:
 - Configuration space: $\{S_i^z = \pm 1\}$
 - Boltzmann weight: $e^{-\beta E[\{S^z\}]}$
- **This lecture:** classical MC algorithms (Metropolis, SW, Wolff)
- **Later lectures:** apply to Kondo-Ising; extend to quantum MC



Monte Carlo Sampling: The Core Idea

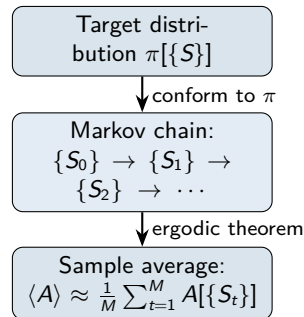
- **Goal:** compute thermal expectation values

$$\langle A \rangle = \sum_{\{S\}} A[\{S\}] \pi[\{S\}]$$

- Boltzmann distribution:

$$\pi[\{S\}] = \frac{e^{-\beta \mathcal{H}[\{S\}]}}{Z}, \quad Z = \sum_{\{S\}} e^{-\beta \mathcal{H}[\{S\}]}$$

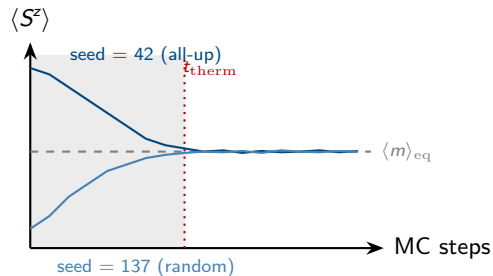
- **Problem:** Z is intractable — combinatorial explosion of configurations
- **Solution:** construct a Markov chain that visits configurations with frequency $\propto \pi$



A Word on Computer Random Numbers

- Computers generate **pseudo-random** numbers — deterministic sequences determined by a **seed**
- Advantage: fully reproducible simulations
- Caveat: long-range correlations may introduce bias
- Different seeds $\rightarrow$ different trajectories through configuration space

Whether an MC algorithm is stable and how to characterize its trajectory will be discussed in Lecture 2.



Two Conditions for Correct Sampling

A Markov chain sampler must satisfy two conditions to converge to the target distribution π :

1. Detailed Balance

$$W(A \rightarrow B) \pi(A) = W(B \rightarrow A) \pi(B)$$

Equivalently: $\frac{W(A \rightarrow B)}{W(B \rightarrow A)} = e^{-\beta(E_B - E_A)}$
 $\Rightarrow \pi$ is a **stationary distribution** of the chain.

2. Ergodicity

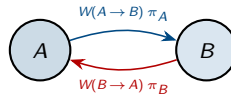
Every configuration is reachable from any other in finitely many steps:

$$\forall A, B: \exists n \geq 1, \text{Prob}(A \xrightarrow{n} B) > 0$$

$\Rightarrow$ the stationary distribution is **unique**.

Why They Suffice

- 1 Detailed balance $\Rightarrow \pi$ is a **fixed point**
- 2 Ergodicity $\Rightarrow$ fixed point is **unique**
- 3 Together $\Rightarrow$ chain converges to π from *any* starting state



Net flow = 0 at equilibrium

Thermalization: When Can We Start Measuring?

Thermalization (Burn-in)

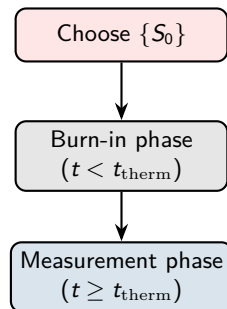
The Markov chain needs t_{therm} steps before samples follow π . Discard the first t_{therm} samples before computing observables.

Practical checks:

- 1 Start from *different* initial configurations (e.g. all spins up vs. random — see previous slide)
- 2 Track $\langle m \rangle$, $\langle E \rangle$, ... until they converge
- 3 Confirm that **independent** runs agree

Advanced: the *algorithmic overlap* q between two replicas provides a thermodynamic-level indicator of equilibration.

Ref: arXiv:2604.10254; detailed discussion in Sec. 7.



Discard burn-in data;
measure from here

The 2D Ising Model: Testbed for Classical Algorithms

Hamiltonian

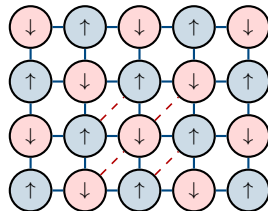
$$\mathcal{H} = -J \sum_{\langle ij \rangle} S_i^z S_j^z$$

- $S_i^z = \pm 1$ on a 2D square lattice
- $J > 0$: ferromagnetic; $J < 0$: antiferromagnetic

With next-nearest-neighbor frustration

$$\mathcal{H} = -J \sum_{\langle ij \rangle} S_i^z S_j^z + J' \sum_{\langle\langle ik \rangle\rangle} S_i^z S_k^z$$

$J' > 0$: competing AFM interaction $\Rightarrow$ frustration



NN: J (solid)

NNN: J' (dashed)

Metropolis Algorithm: Single-Spin-Flip

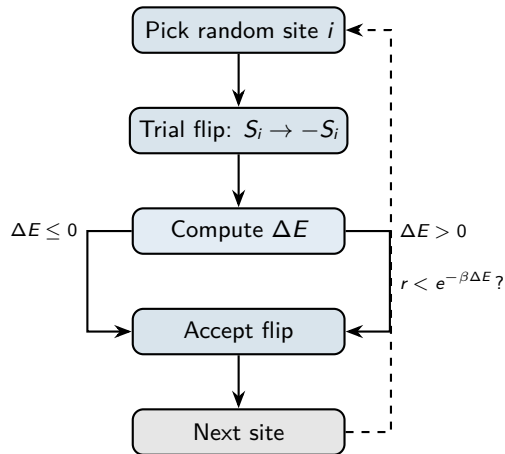
Acceptance Probability

$$W(S \rightarrow S') = \min(1, e^{-\beta \Delta E})$$

where $\Delta E = E(S') - E(S)$ is the energy change from flipping one spin.

Properties:

- **Local** update: flip one spin at a time
- Downhill moves ($\Delta E < 0$): always accepted
- Uphill moves: accepted with probability $e^{-\beta \Delta E}$
- Ergodic: any configuration reachable by sequential flips



Metropolis: Detailed Balance Verification

Proof of Detailed Balance

For a single spin flip at site i , let S' differ from S only at i :

$$\frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = \frac{\min(1, e^{-\beta(E_{S'} - E_S)})}{\min(1, e^{-\beta(E_S - E_{S'})})}$$

Case 1: $E_{S'} \leq E_S \Rightarrow$ numerator = 1, denominator = $e^{-\beta(E_S - E_{S'})}$

$$\therefore \frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = e^{-\beta(E_{S'} - E_S)} = \frac{\pi(S')}{\pi(S)} \quad \checkmark$$

Case 2: $E_{S'} > E_S \Rightarrow$ same result by symmetry.

Ergodicity

Since each spin can be flipped individually, any configuration $\{S\}$ can be transformed into any other $\{S'\}$ by at most N single-spin flips. $\checkmark$

Swendsen–Wang Algorithm: Bond Percolation

Fortuin–Kasteleyn Bond Activation

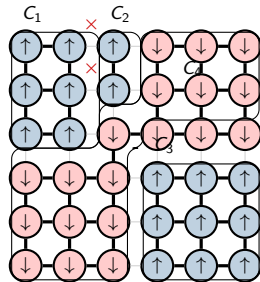
For each pair of **aligned** nearest neighbors $S_i = S_j$:

$$p_{\text{bond}} = 1 - e^{-2\beta J}$$

Activate bond with probability p_{bond} ; never bond anti-aligned pairs.

Algorithm:

- 1 Scan all NN bonds; activate with prob p_{bond}
- 2 Identify connected clusters via union-find
- 3 Flip each cluster independently with prob $1/2$



Swendsen-Wang Flipping Process

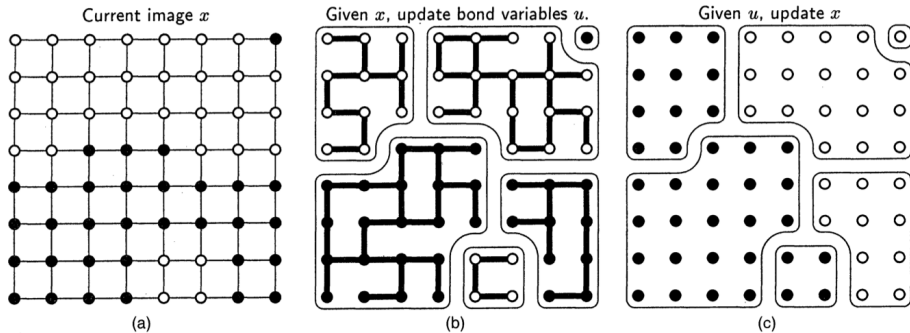


Figure 2. The Swendsen–Wang Algorithm for the Ising Model on the 8×8 Lattice. (a) Initial image x and Markov random field graph for $\pi(x)$. (b) Given the current image x , the bond variables u are generated uniformly over the interval $(0, e^{\beta I[x_i=x_j]})$. If $u_{ij} > 1$ (marked by the thick lines), x_i is constrained to equal x_j . These constraints partition the image into clusters of like-colored sites. Clusters induced by this realization of $u|x$ are outlined. The Markov random field graph for $x|u$ differs from that of x , marginally; the auxiliary vector u strengthens the dependence between some neighboring sites, while completely removing it from others. (c) Given the bond variables u , x is now a coarse image of independent clusters. Because there is no external field in this example, each cluster is recolored black or white with probability .5.

Swendsen–Wang: Detailed Balance

Transition Probability

Probability to go from S to a specific S' :

$$W_{\text{SW}}(S \rightarrow S') = \sum_{B \sim S, S'} P(B | S) \left(\frac{1}{2}\right)^{N_C(B)}$$

Sum over bond configs B compatible with both S and S' ; each of N_C clusters flipped independently with prob $1/2$.

Detailed Balance Holds

In the FK representation ($e^{\beta J|E|}$ cancels in the ratio):

$$\frac{W_{\text{SW}}(S \rightarrow S')}{W_{\text{SW}}(S' \rightarrow S)} = \frac{\prod_{\langle ij \rangle} e^{\beta J(1+S'_i S'_j)}}{\prod_{\langle ij \rangle} e^{\beta J(1+S_i S_j)}} = e^{-\beta(E_{S'} - E_S)} = \frac{\pi(S')}{\pi(S)} \quad \checkmark$$

Key: $p = 1 - e^{-2\beta J}$ is chosen so that $e^{\beta J}(1 - p) = e^{-\beta J}$.

Ergodicity: with prob $(1-p)^{|E|} > 0$, all bonds absent $\Rightarrow N$ single-site clusters $\Rightarrow$ any config reachable. $\checkmark$

SW Detailed Balance: Full Derivation (*backup*)

Step 1. Bond probability:

$$P(B | S) = \prod_{\langle ij \rangle: S_i=S_j} p^{b_{ij}} (1-p)^{1-b_{ij}}$$

$b_{ij} \in \{0, 1\}$: bond indicator in B ; bonds only between aligned spins.

Step 2. Boltzmann weight (FK form):

$$\pi(S) \propto \prod_{\langle ij \rangle} e^{\beta J(1+S_i S_j)}$$

$\Rightarrow e^{2\beta J}$ for aligned, 1 for anti-aligned.

Step 3. Transition probability:

$$W_{\text{SW}}(S \rightarrow S') = \sum_{B \sim S, S'} P(B | S) \left(\frac{1}{2}\right)^{N_c}$$

Step 4. Detailed balance via key lemma
(next slide: $\pi(S) P(B|S) = \pi(S') P(B|S')$):

$$\begin{aligned} \pi(S) W(S \rightarrow S') &= \sum_B \underbrace{\pi(S) P(B|S)}_{= \pi(S') P(B|S')} (1/2)^{N_c} \\ &= \pi(S') W(S' \rightarrow S) \end{aligned}$$

$$\Rightarrow W(S \rightarrow S') / W(S' \rightarrow S) = \pi(S') / \pi(S) \checkmark$$

SW Detailed Balance: Why It Really Works (*backup*)

Key lemma. Define $f(S, B) = \pi(S) \cdot P(B \mid S)$:

$$f(S, B) \propto \prod_{\langle ij \rangle} e^{\beta J S_i S_j} \cdot p^{b_{ij}} (1-p)^{1-b_{ij}}$$

Per-bond contribution to f :

$$\begin{cases} e^{\beta J} \cdot p & b_{ij} = 1, S_i = S_j \\ e^{\beta J} (1-p) \stackrel{!}{=} e^{-\beta J} & b_{ij} = 0, S_i = S_j \\ e^{-\beta J} & b_{ij} = 0, S_i \neq S_j \end{cases}$$

$$e^{\beta J} (1-p) = e^{-\beta J} \text{ iff } p = 1 - e^{-2\beta J} \checkmark$$

Cluster flip invariance.

When cluster C_k flips ($S_i \rightarrow -S_i$ for $i \in C_k$):

- **Intra-cluster:** both spins flip $\Rightarrow S_i S_j$ unchanged $\Rightarrow$ contribution unchanged
- **Boundary** ($b_{ij}=0$, one site in C_k): $S_i S_j$ flips sign, but aligned $\leftrightarrow$ anti-aligned both give $e^{-\beta J} \Rightarrow$ unchanged

$\therefore f(S, B) = f(S', B)$ for every compatible B .

Detailed balance:

$$\begin{aligned} \pi(S) W(S \rightarrow S') &= \sum_B f(S, B) \left(\frac{1}{2}\right)^{N_c} \\ &= \sum_B f(S', B) \left(\frac{1}{2}\right)^{N_c} = \pi(S') W(S' \rightarrow S) \end{aligned}$$

Wolff Algorithm: Single-Cluster Update

Core Idea

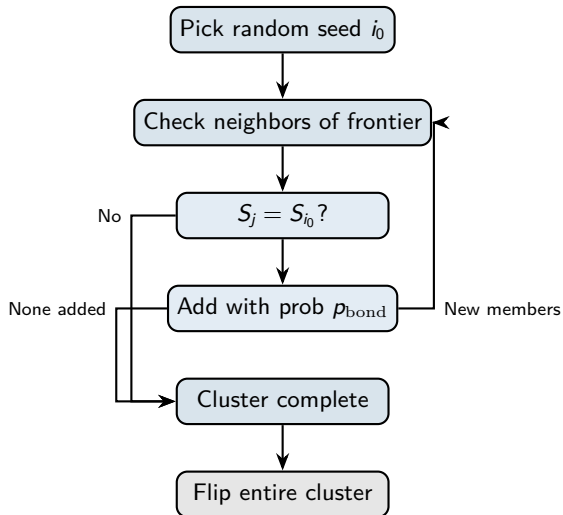
Instead of building *all* clusters (SW), grow **one cluster** from a random seed site using the same FK bond probability:

$$p_{\text{bond}} = 1 - e^{-2\beta J}$$

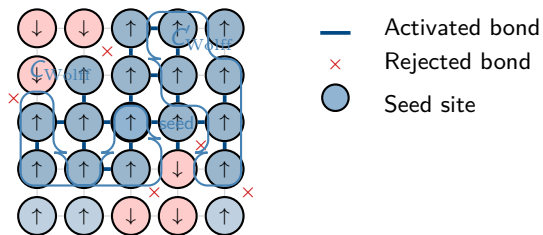
Algorithm:

- 1 Pick a random seed site i_0
- 2 For each unvisited neighbor j of the cluster frontier:
if $S_j = S_{i_0}$, add j with prob p_{bond}
- 3 Repeat step 2 until no new members
- 4 Flip the entire cluster

Ref: U. Wolff, PRL **62**, 361 (1989)



Wolff: Cluster Growth Example



Wolff: Detailed Balance

Transition Probability

$W(S \rightarrow S') \neq 0$ only when S' is obtained by flipping **one** connected cluster C of aligned spins. For such a pair:

$$W_{\text{Wolff}}(S \rightarrow S') = \frac{|C|}{N} \prod_{\mathcal{I}(C)} p \prod_{\substack{\partial C \\ S_i = S_j}} (1 - p)$$

$|C|/N$: seed in C ; $\mathcal{I}(C)$: internal bonds activated; ∂C : boundary bonds rejected (stop growth).

Detailed Balance

$|C|/N$ and $\prod_{\mathcal{I}} p$ cancel in ratio (same cluster, same internal bonds). Only ∂C contributes:

$$\frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = \prod_{\substack{\langle ij \rangle \in \partial C \\ S_i = S_j}} \frac{(1 - p)}{1} \cdot \prod_{\substack{\langle ij \rangle \in \partial C \\ S_i \neq S_j}} \frac{1}{(1 - p)}$$

After flip: aligned $\leftrightarrow$ anti-aligned swap on ∂C . With $(1 - p) = e^{-2\beta J}$: $\frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = e^{-\beta(E_{S'} - E_S)} = \frac{\pi(S')}{\pi(S)} \checkmark$

Ergodicity: cluster of size 1 (all bonds reject) $\Rightarrow$ single spin flip. $\checkmark$

Wolff Detailed Balance: Full Derivation (*backup*)

Key difference from SW: one Wolff step flips only *one* cluster C . $W(S \rightarrow S') = 0$ unless S' differs from S by flipping a single connected cluster. Arbitrary $S \rightarrow S'$ may need *multiple* Wolff steps — but detailed balance is checked per step.

Step 1. For S' reachable via cluster C :

$$W(S \rightarrow S') = \frac{|C|}{N} \prod_{\mathcal{I}(C)} p \prod_{\substack{\langle ij \rangle \in \partial C \\ S_i = S_j}} (1-p)$$

$\frac{|C|}{N}$: seed in C ; $\mathcal{I}(C)$: activated; ∂C : rejected.

Step 2. Cancel common factors:

$$\frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = \frac{\prod_{\partial, S_i = S_j} (1-p)}{\prod_{\partial, S'_i = S'_j} (1-p)}$$

$|C|/N$ cancels (same C); $\prod_{\mathcal{I}} p$ cancels.

Step 3. Boundary bond contributions:

Each boundary bond ($i \in C, j \notin C$) contributes differently:

Bond in S	$W(S \rightarrow S')$	In S'	$W(S' \rightarrow S)$
Aligned	$(1-p)$ (reject)	Anti-aligned	1 (no check)
Anti-aligned	1 (no check)	Aligned	$(1-p)$ (reject)

After flip: aligned $\leftrightarrow$ anti-aligned swap $\Rightarrow$ contributions also swap.

Wolff: Verify Detailed Balance (*backup, cont.*)

Step 4. Substitute $1 - p = e^{-2\beta J}$:

$$\frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = \prod_{\substack{\partial C \\ S_i = S_j}} e^{-2\beta J} \cdot \prod_{\substack{\partial C \\ S_i \neq S_j}} e^{2\beta J}$$

Let $n_{\uparrow}$ = aligned boundary bonds, $n_{\downarrow}$ = anti-aligned:

$$= (e^{-2\beta J})^{n_{\uparrow}} (e^{2\beta J})^{n_{\downarrow}} = e^{-2\beta J(n_{\uparrow} - n_{\downarrow})}$$

Step 5. Energy change on ∂C :

$$E_S|_{\partial C} = -J n_{\uparrow} + J n_{\downarrow}$$

$$E_{S'}|_{\partial C} = -J n_{\downarrow} + J n_{\uparrow}$$

(aligned $\leftrightarrow$ anti-aligned swap after flip)

$$\Delta E = E_{S'} - E_S = 2J(n_{\uparrow} - n_{\downarrow})$$

Conclusion

$$e^{-\beta \Delta E} = e^{-2\beta J(n_{\uparrow} - n_{\downarrow})} = \frac{W(S \rightarrow S')}{W(S' \rightarrow S)} \quad \checkmark$$

Comparison: Metropolis vs. SW vs. Wolff

	Metropolis	Swendsen–Wang	Wolff
Update scope	Single spin	All clusters	One cluster
Clusters/sweep	N (trivial)	$\sim N$	1
Dynamic exp. z	≈ 2	≈ 0.35	≈ 0.25
Cost/flip	$O(1)$	$O(N)$	$O(\langle C \rangle)$
Best for	Generic models	Frustrated systems	Ferromagnets

- **Metropolis**: simplest; universal but suffers *critical slowing down* ($z \approx 2$)
- **SW**: global update via FK percolation; eliminates most critical slowing down
- **Wolff**: single-cluster variant; even lower z than SW for ferromagnets
- All three satisfy detailed balance & ergodicity
- Near T_c : Wolff is typically the most *efficient per unit cost*

(Section 5 — to be written)

(Section 6 — to be written)

(Section 7 — to be written)